

FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10 | SUB-STRAND 2.1: SIMILARITY AND ENLARGEMENT

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: How can a passport photo be made bigger or smaller without changing its shape, and how do its length, area and cost change when we do so? Achieng is a Grade 10 student in Kisumu. She needs passport photos for KCSE registration and a school ID. Mr Otieno, the photographer, prints her photo as stamp size S (3 cm x 4 cm, KES 30), postcard size P (6 cm x 8 cm, KES 120) and large size L (9 cm x 12 cm). Mr Otieno charges KES 2.50 per cm² of paper. Use the data and the ideas from our 12 lessons to answer each question in the space under it. Show your working, write units (cm, cm², cm³, KES, L), and use the words object, image, similar, LSF, ASF and VSF correctly. Write in full sentences when you are asked to explain.

Part 1: Same shape, different size

Print S is 3 cm x 4 cm with a diagonal of 5 cm. Print P is 6 cm x 8 cm with a diagonal of 10 cm. Mr Otieno also made a photo that is 6 cm x 4 cm by doubling only the width of S. (a) Using numbers, test whether P is similar to S. (b) Test whether the 6 cm x 4 cm photo is similar to S. (c) Explain in your own words what 'similar' means, and why Achieng's photo can be printed in different sizes without looking stretched.

Write your answer here:

Part 2: The linear scale factor

(a) Treat S as the object and P as the image. Write the formula for the linear scale factor k and use it to find k for S to P using the width, the height and the diagonal. **(b)** Find k for S to L. **(c)** Mr Otieno wants to turn P back into S. Find k for P to S and say whether this is an enlargement or a reduction. **(d)** Achieng's friend says, 'Mr Otieno made P by adding 3 cm to each side of S.' Is she right? Explain using the lengths of S and P.

Write your answer here:

Part 3: Area and cost

(a) Find the area of S, P and L in cm^2 . (b) Find the cost of each print at KES 2.50 per cm^2 . (c) Compare P with S. By how many times are the lengths multiplied, the area multiplied and the cost multiplied? (d) Explain why the price of P is KES 120 and not KES 60. (e) Without measuring, use $k = 3$ to predict the cost of L compared with S and check it with your answer to (b).

Write your answer here:

Part 4: Beyond photos

(a) A small jerrycan holds 5 L. A similar larger jerrycan has every length twice as big (LSF 2). How much does it hold? Show how you worked it out. (b) A cube has side 2 cm. Find the volume of a similar cube with LSF 3. (c) On a map with scale 1:5,000,000, the road from Nairobi to Mombasa measures 8.8 cm. Find the real distance in km. (d) A classroom plan is drawn to scale 1:50 and measures 16 cm x 12 cm. The real room is 8 m x 6 m. Floor tiles cost KES 800 per m^2 . Find the cost of tiling the floor. (e) Copy and complete: when every length is multiplied by k , the length is multiplied by ____, the area by ____ and the volume by _____. State when this rule works.

Write your answer here:

Part 5: Putting the whole explanation together

Answer the driving question: How can a passport photo be made bigger or smaller without changing its shape, and how do its length, area and cost change when we do so? Write a clear explanation for a classmate who missed the lessons. Use Achieng's prints S, P and L, the 6 cm x 4 cm photo, and the jerrycan or another example. Include (a) the rule for keeping the shape, (b) how length, area and cost change, with numbers, and (c) what goes wrong if only the width is doubled.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Testing and explaining similarity	Checks both equal angles and the same ratio of all corresponding sides with correct numbers, and correctly shows the 6 x 4 photo fails because the ratios are 2 and 1. Explains similar versus congruent accurately.	Checks angles and ratios with mostly correct numbers and identifies the distorted photo as not similar, but the explanation of similar is incomplete.	Uses only looks or one ratio to decide, or does not identify why the 6 x 4 photo is not similar.
Linear scale factor calculations	Uses $k = \text{image/object}$ with corresponding sides and the same units to get $k = 2, 3$ and $1/2$ correctly, classifies the reduction, and explains that multiplying (not adding) keeps the shape.	Calculates most scale factors correctly but gives a weak explanation of multiplying versus adding or of the reduction.	Confuses object and image, inverts the ratio, or uses addition to find the new lengths.
Area, volume and cost reasoning	Correctly finds the areas 12, 48 and 108 cm^2 and the costs KES 30, 120 and 270, and links $\text{ASF} = k^2$ and $\text{VSF} = k^3$ to the cost and the jerrycan, explaining that cost follows area.	Finds the areas and costs correctly and states k^2 and k^3 , but the link between cost and area or the volume reasoning is thin.	Says the cost or area simply doubles, or does not use k^2 or k^3 .
Applying scale to real contexts	Converts the map scale and the scale drawing correctly (440 km, 48 m^2 , KES 38,400) with correct units and shows the	Gets most conversions correct but makes a unit or working slip, or does not show the use of n^2 .	Does not convert units correctly or does not apply the scale to length and area.

	use of n for length and n^2 for area.		
Final explanation and scientific language	Gives a connected, complete answer to the driving question using object, image, similar, LSF, ASF, VSF and Achieng's data, and addresses the stretched photo.	Answers the driving question with mostly correct terms and data but misses one link, for example the stretched photo or the cost.	Gives a list of facts without connecting length, area and cost or without using the key terms.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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